

AI-driven Tree Detection in South Florida: Benchmarks and Fine Tuned models for YOLO and Deep Forest.

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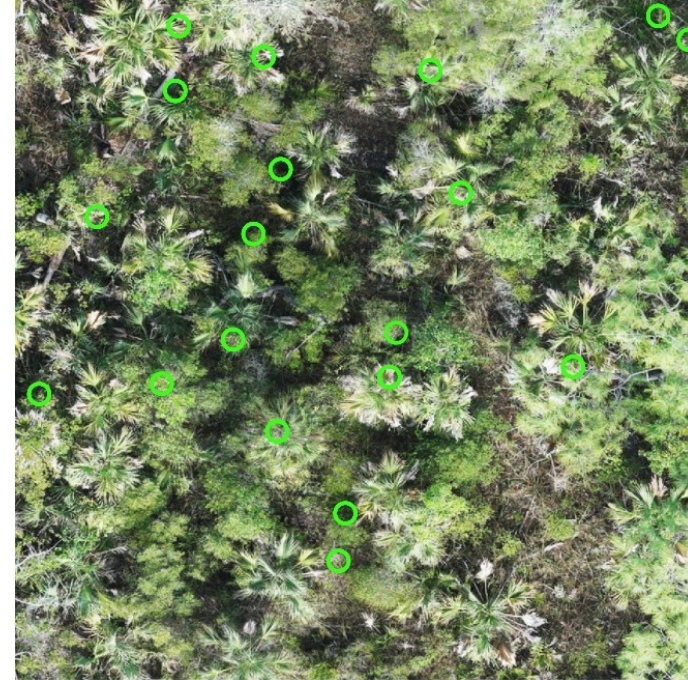
Frost Institute for Data Science and Computing, University of Miami | Summer 2026

My part: one pipeline, two very different forests



UM Coral Gables campus

40 m x 40 m tile.
Sparse, urban, palm-heavy. Inventory of 6,988 confirmed trees.

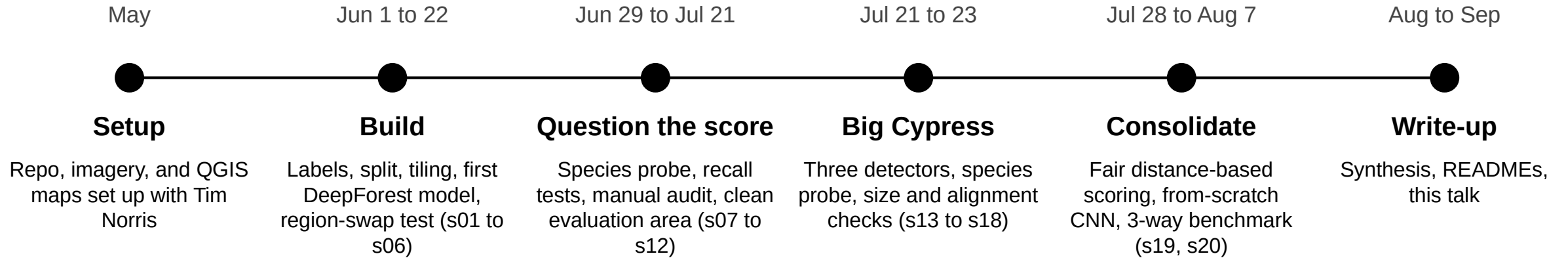


Big Cypress National Preserve

20 m x 20 m patch.
Dense wetland, crowns overlap. About 1,500 to 2,600 trees per hectare. (Circles = surveyed palms.)

The question: can a model find every individual tree in an ordinary color (RGB) photo, and can we trust the score it gets?

Timeline: May to September



How it was built

20

numbered scripts, each one answering one question

25

commits, each a finished experiment with its results

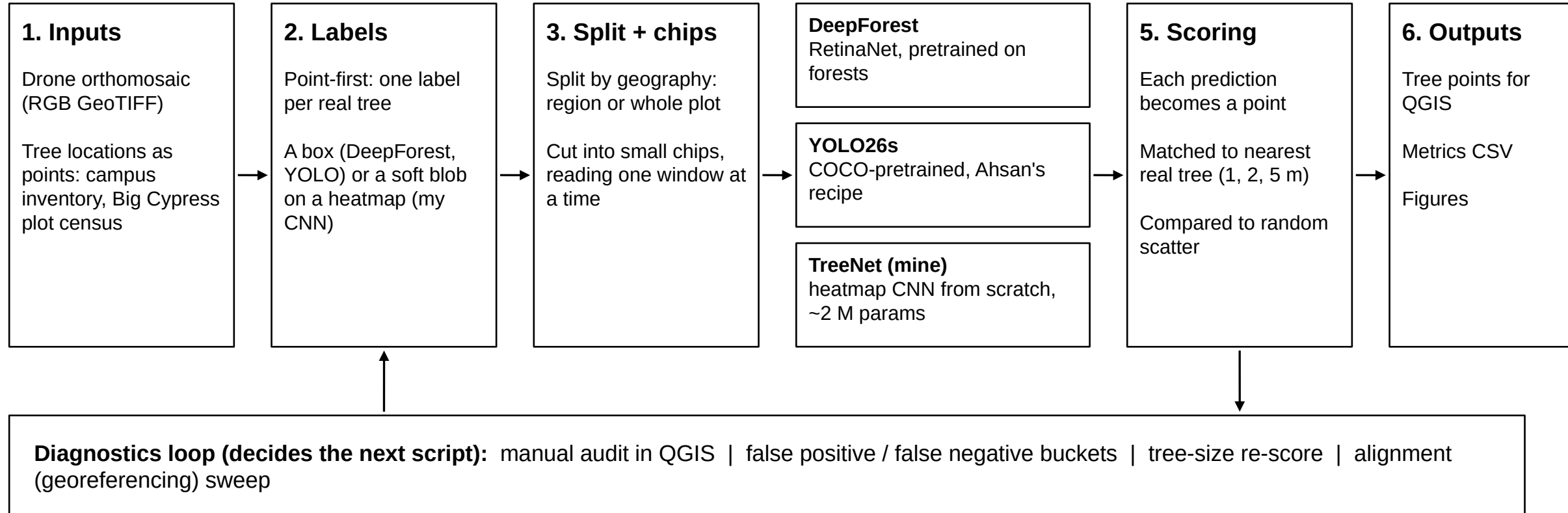
Every step

saves a figure and a CSV, so results get checked by eye

1 laptop

all training on a MacBook (M3 Pro, 18 GB, Apple GPU)

Architecture: from drone photo to tree map



config.yaml
every setting in one file

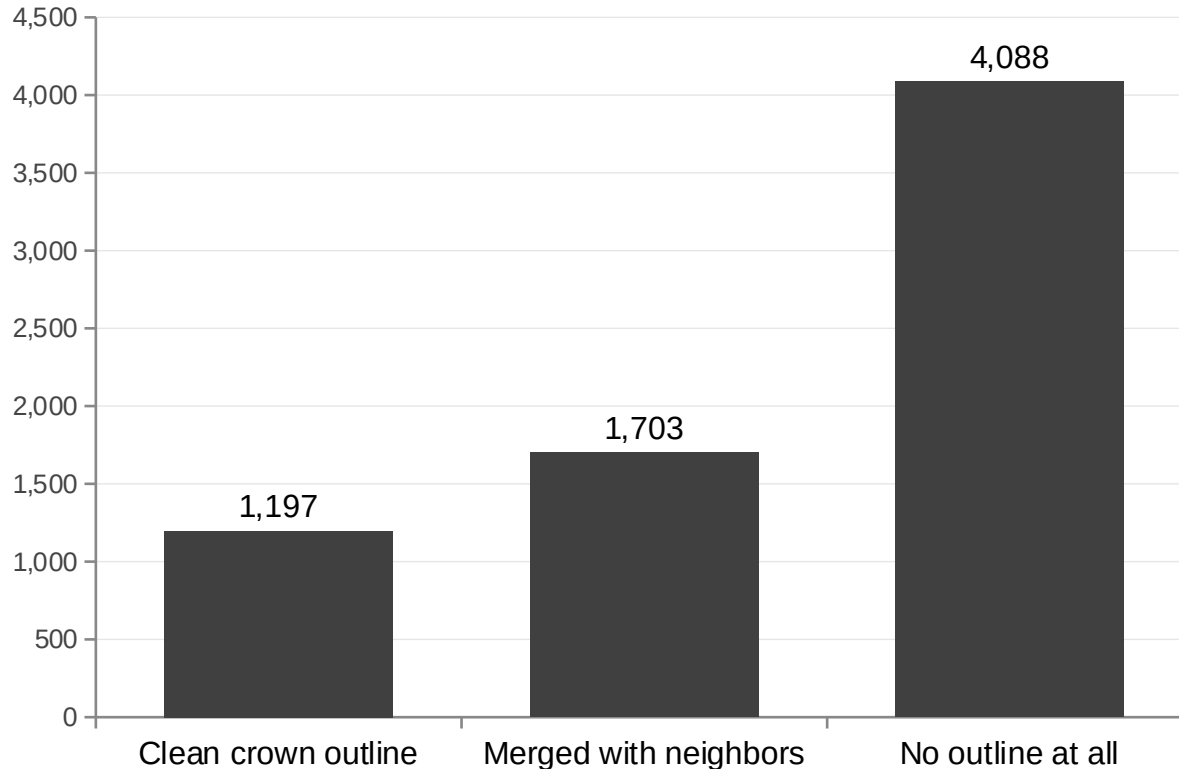
SOFLO_DATA_ROOT
~3.7 GB of imagery stays off GitHub

PyTorch on Apple GPU
windowed reads, peak RAM under 6 GB

Git
one script + one commit per experiment

Step 1: the old labels missed most of the trees

6,988 confirmed trees vs. the prior model's crown outlines



~6 in 10

confirmed trees had no outline from the prior model

640

outlines merged two or more trees into one crown

Design choice: trust the points, not the outlines

Every confirmed tree gets exactly one box: its real crown where a clean outline exists, otherwise a ~7 m box sized from the data. Each box is tagged with where it came from.

Step 2: a first model, and a score that looked bad



Blue = train region (1,951 trees), red = test region (2,211 trees)

F1 = 0.39

precision 0.40, recall 0.39, scored by box overlap (IoU 0.4)

Model: DeepForest (RetinaNet pretrained on forest canopy), fine-tuned on campus tiles

Design choice: split by geography, not random trees, so the test area is ground the model never saw

Training: early stopped at epoch 13 (best epoch 5), about 28 minutes on the laptop

Swap test: trading the train and test regions moved F1 from 0.39 to 0.32, so the score depends on where you test

Next move: not a bigger model, but checking whether the score itself was right

Step 3: was the model wrong, or the answer key?

The audit: for every red box the model drew, I opened the imagery and asked "is there a real tree in here?"

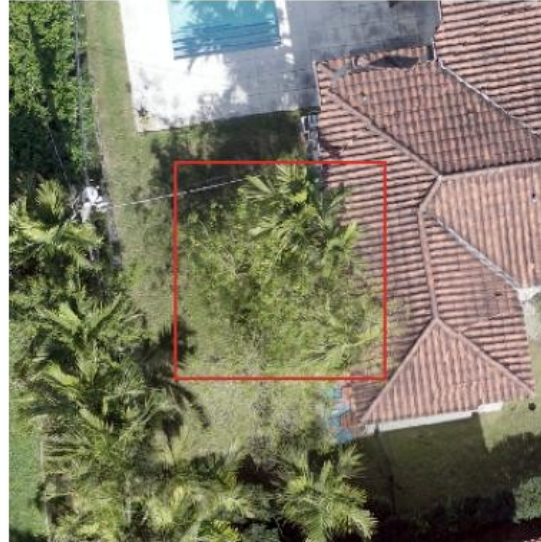
**Real tree
not in the inventory**



**Real tree
not in the inventory**



**Real trees
one box over several crowns**



**A real error
bush, not a tree**



4 of the 100 random "false positives" I checked one by one in QGIS. Red box = a tree the model found that the inventory has no record of.

94 of 100

were real trees the inventory never recorded (3 were real errors, 3 unsure)

5.4%

of misses were truly undetected; the rest had a box on or near the tree, just too loose

~0.61, not 0.96

only 43 of the 94 were new trees (51 were loose boxes), so extrapolated precision is ~0.61

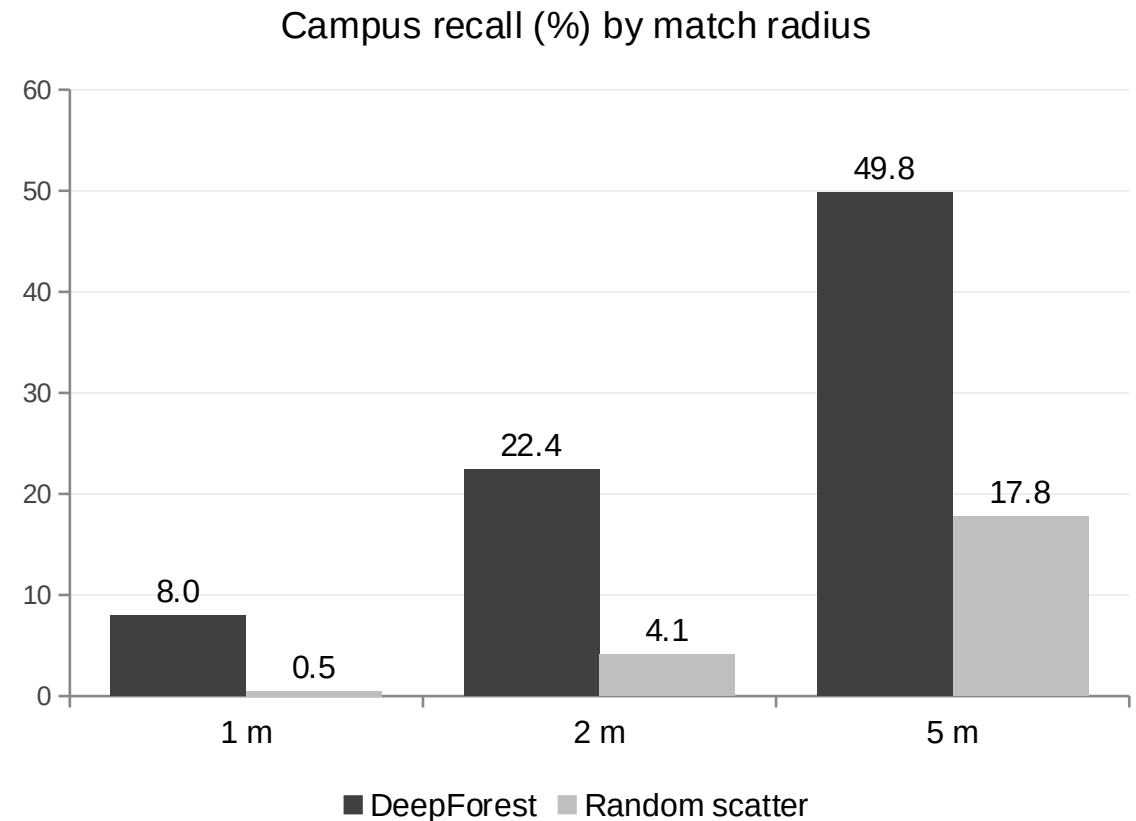
Step 4: change the ruler, and always compare to random

Problem: box overlap punished loose boxes and trees missing from the inventory

New ruler: turn each prediction into a point and match it to the nearest real tree within 1, 2, or 5 m, one match per tree

Random floor: scatter the same number of random points and score them the same way. A model only counts if it beats that.

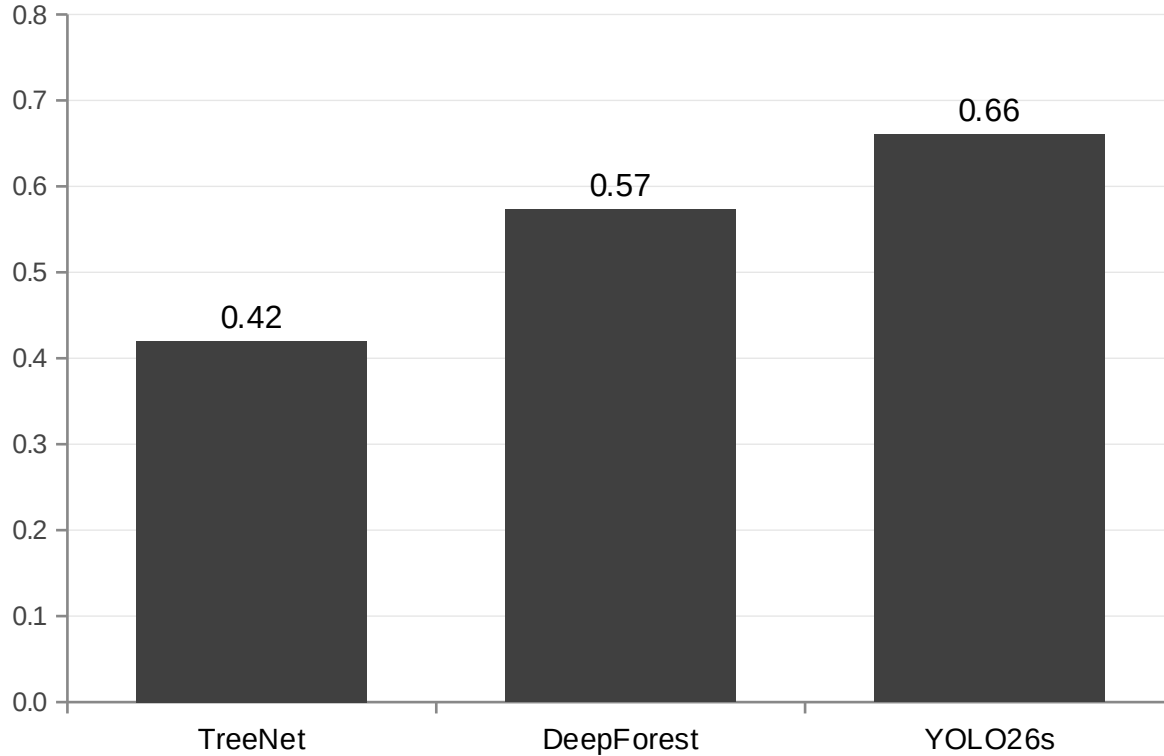
Fair region: score inside a 250 m x 250 m labeled area so unlabeled ground doesn't count against the model



Campus at 5 m: F1 0.57, and the model finds 32 points more of the trees than random does

Campus outcomes: three detectors, one fair comparison

F1 at 5 m, same data, split, and scoring



TreeNet = mine, from scratch | DeepForest = mine, fine-tuned | YOLO26s = Ahsan
Precision / recall: TreeNet 0.37 / 0.49 | DeepForest 0.67 / 0.50 | YOLO26s 0.65 / 0.67

Detection

- My from-scratch model finds almost as many trees as DeepForest (recall 0.49 vs 0.50); the gap is false alarms
- Read as an experiment: pretraining is worth about 0.15 to 0.24 F1 on this site

Species

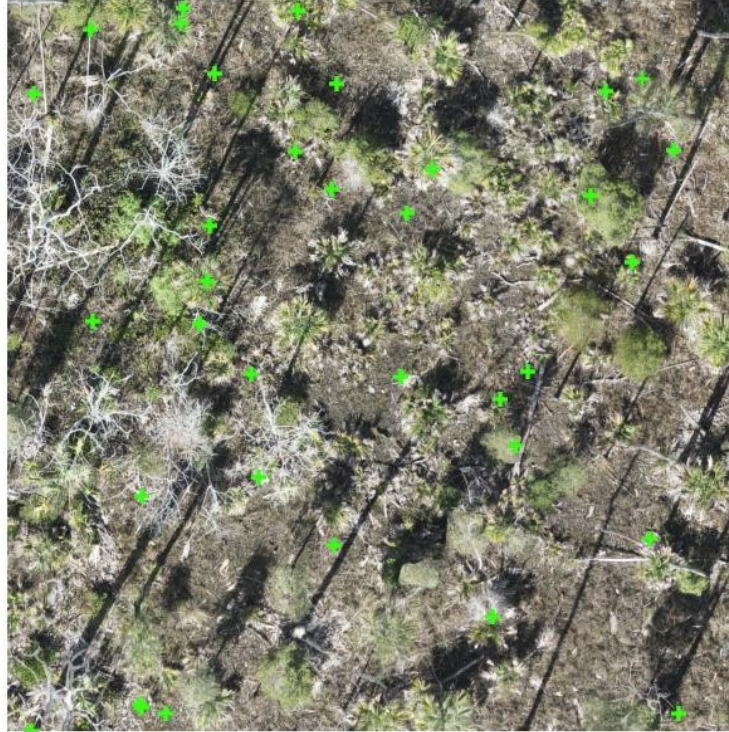
- Palm vs. not-palm on labeled tree crops: 89.9% accuracy, vs. 54.6% by always guessing the majority (an upper bound)
- Pine and cypress couldn't be trained: 55 pine and 22 cypress vs. 3,461 other trees (157 to 1)

Big Cypress: same playbook, hit a wall

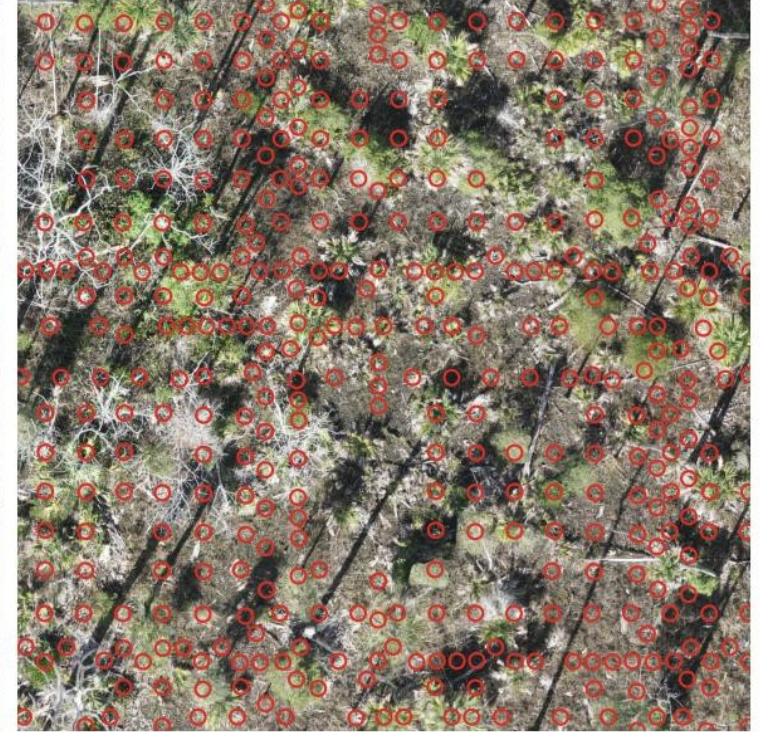
The drone photo
30 m x 30 m of Big Cypress



What the ground survey says
171 surveyed trees in this window



What the model returns
403 detections in the same window



Held-out plot, one 30 m window three ways. Green + = surveyed tree, red circle = model detection.

Whole plot: 2,576 surveyed trees, 4,212 detections. Every detector lands at F1 ~0.11 at a 1 m radius (random scatter gets ~0.10); DeepForest boxes found 3.9% of trees.

Species too: pine / cypress / palm reached 56.9%, worse than always guessing pine (72.1%)

Beating random by recall:
campus +32 points (at 5 m)
Big Cypress +2 points (at 1 m)

Ruling out the easy excuses

Architecture

Three model families
(boxes, heatmap, YOLO)
all land at F1 ~0.11

Training

Weighted MSE loss
collapsed to a blank map;
focal loss learned, still
~0.11

Resolution

Tried 1.6, 3, 5, and 10 cm
per pixel; 5 cm best, none
fixes it

Box size

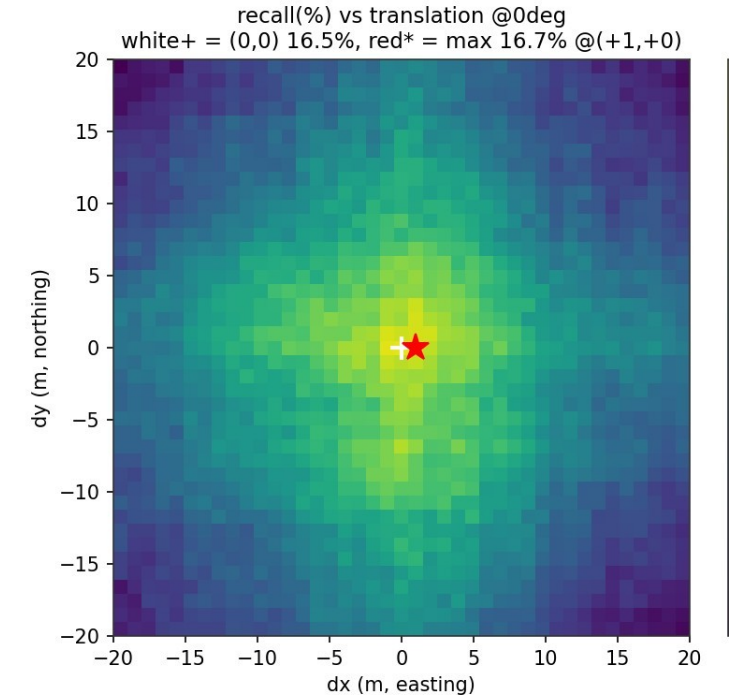
Swept 1.5 to 3 m; 2 m best,
every size near random

Tree visibility

Recall stays ~17% even for
the biggest 10% of trees

Alignment

Palm points sit on palm
crowns; shifting or rotating
the survey never helps



Recall vs. shifting the survey points up to 20 m: best
at no shift

Conclusion: flat RGB can't separate overlapping crowns in dense wetland. Height data is the lever, not a better model.

Where I used AI, and where I didn't

AI-assisted (Claude, mostly Opus 4.8)

- Drafting and debugging the pipeline scripts
- Plotting and figure code for every step
- READMEs, methods notes, and commit write-ups
- Writing the Claude Code prompts, with me setting the direction (next slides)

Done myself

Data inspection: finding the missing-tree problem and choosing point-first labels

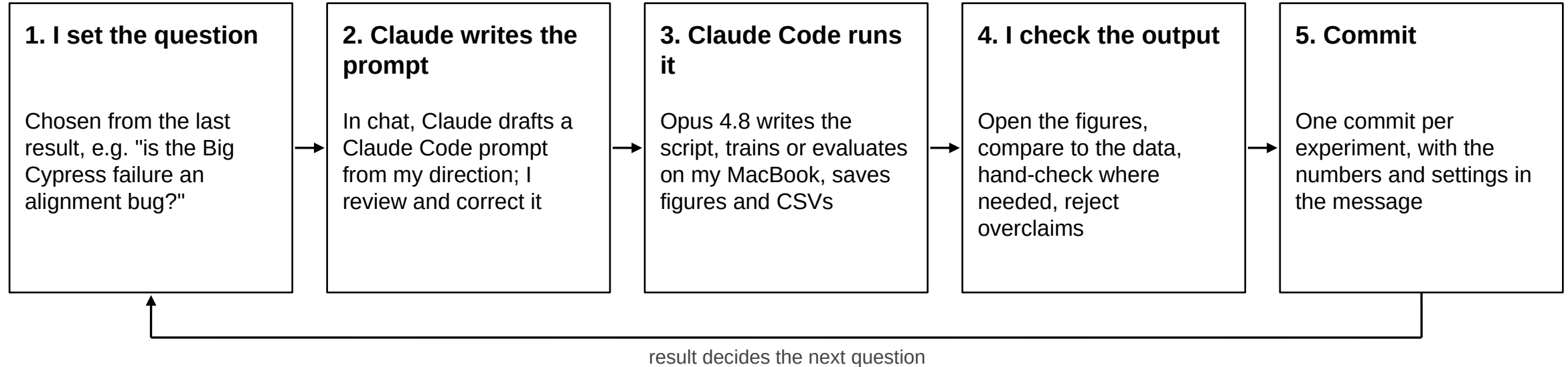
Experiment design: what to test next, which excuses to rule out, when to stop

Manual audit: hand-tagging 100 false positives in QGIS, the palm visual check

Interpretation: deciding what's real vs. an artifact, and catching overclaims (0.96 became ~0.61)

AI made the building fast. Every conclusion was checked against the data by me.

AI reproducibility: how each experiment was run



Models and tools

- Claude (chat) wrote every Claude Code prompt, guided by me
- Claude Opus 4.8 in Claude Code for most experiments and training runs
- The last commit (s20) credits Claude Fable 5 instead
- Every run executed locally on my MacBook, not in the cloud

What makes it reproducible

- Every setting in config.yaml, fixed random seeds
- Exact numbers and run settings written into each commit
- Commits from July on link to the Claude session that produced them
- Baseline check: s09 re-runs the saved model and stops if scores drift by more than 0.03

What a prompt looked like

Example: the s18 alignment check (the spec recorded at the top of the script, condensed)

QUESTION

Is the Big Cypress census-to-imagery alignment offset, rotated, or flipped?

CONSTRAINTS

No retraining, no model changes. Pure diagnostics. Reuse the cached s16 predictions.

STEPS

1. Palm test: 20 m crops with only palm points over the imagery
2. Sweep shifts (+/-20 m), rotations (+/-20 deg), and flips; report the best one
3. Check plots and census are axis-aligned 100 m squares
4. Plain verdict: correct / offset / rotated / flipped

OUTPUTS

Write only under outputs/bcnp_georef and reports/bcnp_georef. Figures + CSV.

The template every prompt followed

- 1 **One question:** a single thing to find out per script
- 2 **Hard constraints:** what not to touch: no retraining, keep the test set out
- 3 **Staged steps:** so partial results survive a crash
- 4 **Laptop limits:** windowed reads, small batches, 18 GB of memory
- 5 **Named outputs:** its own folders, figures, and CSVs
- 6 **A plain verdict:** which I then checked against the figures

My part in every prompt: picking the question and the constraints, and pushing back when a verdict overclaimed.

What I learned

1

Check the answer key first

94 of 100 "mistakes" were real trees the inventory missed.

2

The metric is a design choice

Same model: F1 0.39 by box overlap, 0.52 by distance on the same test area.

3

Always report a random baseline

In dense forest it showed a decent-looking 19% recall was chance.

4

A negative result counts

Ruling out six causes turned "it failed" into "RGB alone can't do this."

5

Data and pretraining beat clever models

Pretraining was worth 0.15 to 0.24 F1; no model fixed Big Cypress.

6

Small steps keep it honest

One script, one question, one commit made every result traceable.

Next steps

Height data for Big Cypress: request the canopy height model from the drone processing (Tim Norris)

Fix the data: re-export three broken plot images (plot_8_3, plot_11_2, plot_12_3)

Finish the clean evaluation area: hand-label every tree in the 250 m square for a fully trustworthy campus score

Better model shape: crown segmentation or height-aware models instead of flat boxes

Cleaner evaluation: a separate validation region so the test set never picks the checkpoint

Write-up: research summary of both sites

Thank you

Up next: Ahsan, on the YOLO26s campus model

Code: soflo-tree-models (rayan branch)